

# AI and Meat Industry Water Usage: A Good Faith Discussion

Posted by u/Brilliant\_Goat\_2361 • • 0 points • 16 comments

I want the preface this by saying I do not intend to point at imagined hypocrisy and undermine the anti-AI argument. This is just me expressing my thoughts, comparing the two industries, and not me giving validity to disingenuous AI-brained arguments.

The AI and meat industries are both extremely bad for the environment and for people. They both require vast amounts of water to operate, but the rest of their negative effects are different.

The meat industry takes up vast swathes of land and it is one of the primary reasons why the Amazon Rainforest is being destroyed, to clear the land for the beef industry.

Industrialized animal agriculture is also a breeding ground for new pathogens as livestock are given a constant stream of antibiotics so that they don't succumb to the diseases that are transmitted in the inhumane environments they are housed in. These environments eventually create antibiotic resistant bacteria, which could spread to humans and cause another pandemic. The meat industry contributes about [5.9% of greenhouse gases] (<https://www.wri.org/insights/4-charts-explain-greenhouse-gas-emissions-countries-and-sectors>) (old data, probably more of a percentage today), which are accelerating climate change. Also, there is [ongoing research] (<https://pmc.ncbi.nlm.nih.gov/articles/PMC6971786/#abstract1>) suggesting that too much red meat can cause serious health problems.

While the meat industry is bad in its own right, but I believe that AI is still worse. AI data centers consume [massive amounts of energy] (<https://www.technologyreview.com/2025/05/20/1116327/ai-energy-usage-climate-footprint-big-tech/>) – which is mostly created by burning fossil fuels. This is problematic because corporations keep building more and more data centers, hungry for fossil fuel power, in an era where climate has passed the boiling point and is already wreaking havoc. I already mentioned the [water usage,] (<https://cee.illinois.edu/news/AIs-Challenging-Waters>) but many of these data centers are [built in areas where there is preexisting water scarcity] (<https://dgtlinfra.com/data-center-water-usage/>) and are actively draining the local water supply.

These issues are bad, but the worst part about AI might be the social alienation and lobotomizing of human intelligence. There are a host of studies that indicate that Large

Language Models [erode critical thinking skills]([https://phys.org/news/2025-01-ai-linked-eroding-critical-skills.html#google\\_vignette](https://phys.org/news/2025-01-ai-linked-eroding-critical-skills.html#google_vignette)) via [cognitive offloading] (<https://www.mdpi.com/2075-4698/15/1/6>). In normal words: [AI makes people stupid] (<https://www.media.mit.edu/projects/your-brain-on-chatgpt/overview/>) because they get the machine to do the thinking for them. This is a real problem because our high tech society relies on people actually knowing how our technology works; if people don't know how the machines of society work, then society collapses.

AI chatbot use [weakens people's social skills] (<https://www.psychologytoday.com/us/blog/urban-survival/202410/spending-too-much-time-with-ai-could-worsen-social-skills>), separates them from real human interaction, and creates a population of vulnerable people who are [dependent on AI] (<https://www.psychologytoday.com/us/blog/urban-survival/202502/how-ai-could-shape-our-relationships-and-social-interactions>). This separation from people makes it much easier for people to lose touch with reality and [develop mental illness] (<https://www.psychologytoday.com/us/blog/dancing-with-the-devil/202506/how-emotional-manipulation-causes-chatgpt-psychosis>). Yes, AI use can create mental illness. There have already been documented cases of people being committed and jailed due to [ChatGPT Psychosis](<https://www.psychologytoday.com/us/blog/dancing-with-the-devil/202506/how-emotional-manipulation-causes-chatgpt-psychosis>).

There are many more reasons why AI is terrible, but I'm getting tired and I feel I have already made my point. The meat industry is bad, but AI is worse. This is why I think our prioritization of anti-ai arguments is valid. Because it will be impossible to fight the multi-billion dollar meat industry and its government lobbyists if the general population is too stupid, complacent, and sedated by AI to organize effectively.

So policies I think would be good for the world would be: the banning of LLMs, enforcing copyright law on generative AI companies (thereby destroying them financially), and removing AI from public schools. AI aggregation and surveillance should also be made illegal as should data brokering. There should also be more social stigma against AI too, because it really is bringing about the death of human civilization.

I also think that we should keep open minds about anti meat industry policies. It's not impossible to be both anti-AI and anti meat industry at the same time. I think that governments, especially the US government, should stop subsidizing the meat industry; it's long past time that it should develop itself in a way that is sustainable. That will mean that there is less meat being produced, that it will be more expensive, but it also means that it will be forced to be higher quality to justify the price. The shrinking of industrialized animal agriculture will help stabilize the environment, and the freed up land could be used for better use, like carbon capturing trees or something. I think that lab grown meat – while still needing more investment and improvement – could be a healthier, more humane and

environmentally friendly alternative to the traditional stuff.

## Comments

**u/Amphy64** • 1 points • 2025-12-18 20:30 UTC

Stumbled on your post while looking for figures, have you seen this?

<https://www.newrootsinstitute.org/articles/violence-translating-outside-the-slaughterhouse>

I'm not interested in defending AI, it just seems unreasonable, and frankly just whitewashing animal ag. to compare what is largely speculation about AI impacts on people to this.

And of course animal agriculture is worse environmentally, there's no comparison.

**u/Brilliant\_Goat\_2361** • 1 points • 2025-12-21 12:53 UTC

Yeah, I've become vegan pilled since this post. Now I am extremely opposed to animal agriculture because of the harm and suffering it is causing every day, and the environmental impact. I was wrong about the water consumption and greenhouse gas emissions from data centers in that industrial agriculture is much worse and will remain so.

I'm still very much against AI; I still think that it is a very bad for a multitude of reasons including, but not limited to the social isolation it creates, the cognitive offloading it incentivizes, and the harm it is causing in the creative, professional, and academic fields.

**u/Amphy64** • 1 points • 2025-12-21 21:32 UTC

Awesome! 💖🌱

Yep, and generative AI can be purely and simply rather a nuisance, as though determining what's real online wasn't bad enough already. Doing fibre crafts, it's shocking seeing how fast it's gone from impossible project images, to ones even experienced crafters can be fooled by, and of course now there are accounts selling fake patterns, taking away from the real creators. So it's obvious why people in those spaces would be frustrated with AI, I just wish for us to focus on solid reasons like that, and be able to talk about animal agriculture like wool production in terms of environmental impacts (a biodiversity disaster, too, sheep eat everything).

**u/funky\_monkey13** • 1 points • 2025-08-18 06:03 UTC

Agriculture feeds people. AI will just make more people starve.

**u/FranklyNotThatSmart** • 1 points • 2025-08-17 11:37 UTC

The meat industry is horrid for emissions particularly beef with methane emissions, land clearing and water usage. It's getting worse and I'm ofc against it- similarly I'm against AI's usage of water, power and land. AI however, has had such a negative socio economic impact on society in the 3 years it's become mainstream through LLMs, img gens and what not who's to say what could happen in future. I think you've really captured a lot of my frustrations about AI and I hope this makes its way to some pro AI people.

u/Artemis\_Platinum • 1 points • 2025-08-17 04:02 UTC

I'm told that Deepseek is a great deal more efficient than most other LLM AI models right now. If I assume that's true I'm kinda hoping that maybe other companies will follow suit and ... significantly reduce the impact, since efficiency directly translates to costing less for these companies to run the damn things.

I'm not offering a defense of AI; I'm just looking for a small victory where we can get them.

u/OfficialHashPanda • 1 points • 2025-08-17 14:51 UTC

The degree to which deepseek is more efficient was very overstated. It did make some innovations regarding training techniques, but many of those were likely already employed by closed-source models.

However, the real problem is that if you can train a model more efficiently, then that just means that you can train even stronger models with more resources. In other words, it won't actually cut down on the total demand for hardware/power/water.

u/printmyplastic • 1 points • 2025-08-16 15:31 UTC

Fact-checking this AI vs Meat Industry comparison - the numbers don't add up

This is a thoughtful post, but several key claims need significant corrections based on 2023-2025 research:

WATER USAGE - Claim verified but scale misleading

You're absolutely right that data centers are built in water-stressed areas. [Bloomberg's 2025 analysis] (<https://www.bloomberg.com/graphics/2025-ai-impacts-data-centers-water-data/>) shows 2/3 of new AI data centers since 2022 are in water-stressed regions. [Google used 8.1 billion gallons in 2024] (<https://blog.google/outreach-initiatives/sustainability/2024-environmental-report/>) (up 88% since 2019), [Meta used 1.29 billion gallons in 2022] (<https://sustainability.atmeta.com/2024-sustainability-report/>). BUT the scale comparison is crucial: Daily meat consumption uses ~31,500x more water than daily AI usage. Your meat consumption (400g daily) = 2,364 liters vs 3 AI queries = 75ml.

ENERGY - Major correction needed

The "mostly fossil fuels" claim is outdated. Current renewable adoption: [Amazon/AWS: 100% renewable matching (2023)] (<https://aws.amazon.com/sustainability/>), [Meta: 100% since 2020] (<https://sustainability.atmeta.com/blog/2024/08/28/meta-launches-2024-sustainability-report/>), [Google: 66% carbon-free with 100% renewable matching since 2017] (<https://blog.google/outreach-initiatives/sustainability/2024-environmental-report/>), [Microsoft: 80% renewable, targeting 100% by 2025] (<https://blogs.microsoft.com/on-the-issues/2024/05/15/microsoft-environmental-sustainability-report-2024/>). [EU livestock sector consumes 43-75 MJ per kg] (<https://www.mdpi.com/2076-3417/12/4/2142>) - the global meat industry uses ~2,000 TWh annually vs AI's estimated 100-300 TWh. Meat industry uses 7-20x more energy than all AI systems combined.

MEAT EMISSIONS - You significantly understated

Your 5.9% figure is wrong. [Latest FAO data: livestock produces 6.2 GtCO<sub>2</sub>eq annually = 12% of global emissions] (<https://www.fao.org/newsroom/detail/new-fao-report-maps-pathways-towards-lower-livestock-emissions/en>). Historical estimates range 14.5-19.6%. You're off by 50-70%. Amazon

deforestation is accurately described - [Brazil's cattle accounts for ~70% of deforestation] (<https://www.farmsanctuary.org/news-stories/amazon-deforestation-beef-industry>), [800M+ trees felled 2017-2022](<https://news.mongabay.com/2023/10/agriculture-in-the-pan-amazon-beef-production-models/>).

## HUMAN HEALTH IMPACTS - The missing piece

You didn't mention the meat industry's direct human death toll: [3.3 million annual deaths from air pollution](<https://www.nature.com/articles/nature15371>) linked to livestock ammonia emissions, [128,000 annual US deaths from red meat consumption](<https://journals.plos.org/plosmedicine/article?id=10.1371/journal.pmed.1000058>), [35,000+ deaths from antibiotic resistance] (<https://www.cdc.gov/antimicrobial-resistance/data-research/threats/index.html>) with livestock antibiotic use as major contributor, [4,300+ infant deaths from CAFO water pollution] (<https://www.pnas.org/doi/10.1073/pnas.1913945117>). Total: 500,000+ annual deaths directly attributable to meat industry vs dozens of documented AI mental health cases.

## COGNITIVE EFFECTS - Mixed evidence

Peer-reviewed support exists but it's nuanced: [Gerlich (2025): significant negative correlation ( $r = -0.49$ ,  $p < 0.001$ ) between frequent AI use and critical thinking](<https://www.mdpi.com/2075-4698/15/1/6>), [Zhai et al. (2024): systematic review found consistent evidence of cognitive offloading effects](<https://slejournal.springeropen.com/articles/10.1186/s40561-024-00316-7>), BUT [Essel et al. (2024): ChatGPT enhanced critical thinking in controlled study] (<https://www.sciencedirect.com/science/article/pii/S2666920X23000772>). Most studies are cross-sectional, not longitudinal.

## "CHATGPT PSYCHOSIS" - Documented but not systematic

Clinical case reports exist: [Østergaard (2023) in Schizophrenia Bulletin discusses AI-induced delusion risk](<https://pmc.ncbi.nlm.nih.gov/articles/PMC10686326/>), [Dr. Keith Sakata (UCSF) reports 12 hospitalizations in 2025](<https://www.psychologytoday.com/us/blog/urban-survival/202507/the-emerging-problem-of-ai-psychosis>). However, no controlled longitudinal studies exist. Causation vs correlation unclear.

## THE SCALE REALITY CHECK

Here's where your argument falls apart: Energy: Meat industry uses 7-20x more energy than all AI systems. Carbon: 1 kg beef = 60,000g CO<sub>2</sub>e vs 1 AI query = 2.2g CO<sub>2</sub>e (27,000x difference). Water: 1 kg beef = 616,000 AI queries in water usage. Deaths: 500,000+ annual deaths from meat industry vs dozens of AI mental health cases. Daily impact: Meat consumption = 1,000-30,000x greater environmental impact than daily AI usage.

## POLICY EFFECTIVENESS

Livestock accounts for 25% of human water use and 18% of global GHG emissions. [Dairy alone uses 250x more water annually than global AI systems](<https://bryantresearch.co.uk/insight-items/comparing-water-footprint-ai/>). [EPA's 2024 rule targets 100M pounds of annual pollution reduction](<https://www.epa.gov/eg/meat-and-poultry-products-effluent-guidelines-2024-proposed-rule>) from meat processing. Meat regulation has decades of proven framework while AI provides massive economic benefits.

## BOTTOM LINE

Your concerns about AI's social impacts deserve discussion, but the environmental and health arguments are backwards. Meat production's impact exceeds AI by orders of magnitude: energy (7-20x), carbon (27,000x per kg), water (616,000x per kg), and human deaths (500,000+ annually vs dozens of cases). A 10% reduction in meat consumption has equivalent climate impact to eliminating personal AI usage entirely, while potentially preventing 50,000+ deaths annually. The cognitive/social concerns need more research, but environmental priority should clearly be meat industry regulation given the massive scale differences.

Additional sources: [MIT 2025 AI environmental analysis](https://news.mit.edu/2025/explained-generative-ai-environmental-impact-0117), [Nature on AI energy consumption] (https://www.nature.com/articles/d41586-025-00616-z), [UNEP AI environmental report] (https://www.unep.org/news-and-stories/story/ai-has-environmental-problem-heres-what-world-can-do-about)

**u/Brilliant\_Goat\_2361** • 1 points • 2025-08-17 15:23 UTC

Ok. Thank you for clarifying. I don't really trust the pledges of corporations to reach carbon neutrality because they have abandoned their pledges in the past. Other than that, everything else seems to be correct. We should still regulate and curtail both industries.

**u/Moth\_LovesLamp** • 6 points • 2025-08-15 21:02 UTC

Just because it's well known Agriculture + Meat Industry is the biggest consumer of water and the biggest environmental damaging in the world, it doesn't undermine the fact that AI Data Centers also consume a lot of water and electricity, and much more rapidly than anything before.

It's like trying to justify the existence of missile launchers with guns. They are both bad and could be used in much more sustainable and ethical manners, but profit and greed speaks louder.

**u/printmyplastic** • 0 points • 2025-08-16 15:35 UTC

>Just because it's well known Agriculture + Meat Industry is the biggest consumer of water and the biggest environmental damaging in the world, it doesn't undermine the fact that AI Data Centers also consume a lot of water and electricity, and much more rapidly than anything before.

I mean, it definitely does seem to undermine it. As an AI using vegan, you people making a big deal about the 2% emissions and localized water use caused by AI use, while mindlessly consuming beef, come off as as unhinged and psychotic, completely detached from reality.

**u/Brilliant\_Goat\_2361** • 1 points • 2025-08-16 13:51 UTC

I agree. I think that the energy and water demands of the agriculture and meat industry are very large (because it is an established industry) and growing steadily, but AI's demands are growing exponentially. Soon the water and energy demands of AI will surpass that of the meat industry, which is really bad because water scarcity is already an issue.

**u/printmyplastic** • 0 points • 2025-08-16 15:36 UTC

>Soon the water and energy demands of AI will surpass that of the meat industry, which is really bad because water scarcity is already an issue.

Citation needed.

**u/AnyVanilla5843** • 0 points • 2025-08-15 22:38 UTC

Your wrong on that. The modern statistics of how much power and water goes towards ai in the datacenters is smaller than the entire meat industry. It's smaller than most other industries. I'm still confused on where you people pull these numbers from. Especially since 4 different parties including non-related ones have proven the numbers u toss around are false.

**u/QuantumModulus** • 2 points • 2025-08-16 04:35 UTC

It's the rate of change that's unprecedented, not the current absolute amount.

**u/TougherThanAsimov** • 6 points • 2025-08-15 21:00 UTC

I know a good bit about livestock production from college, and... \*Yeah, I can't deny the environmental concerns.\* If you need a reason to go vegan, mention that and you need not say more.

The industry actually had to crack down on those additional antibiotics in the feed with some legislation in 2017. Arguably, it probably shouldn't have taken that long though to make those veterinary feed directives did their job. And remember, it's not just about the land use for pasture or housing. Every animal has a feed-to-meat ratio of over one, so that's extra corn/soy getting used \*especially\* for feedlot beef. Cattle have the worst feed conversion ratio and don't eat straight roughage (As in fresh grass or hay) in the feedlots.

If we could make fake meat products cover at least half of our food demand, that would cut down on **\*\*a lot\*\*** of our resource issues while meat breeds of livestock would still have a niche. \*That would be a good deal.\*

**u/printmyplastic** • 1 points • 2025-08-16 15:38 UTC

>If we could make fake meat products cover at least half of our food demand, that would cut down on **\*\*a lot\*\*** of our resource issues while meat breeds of livestock would still have a niche. \*That would be a good deal.\* Heck, I tried a fake meat burger one time because I respect the idea.

The problem is that many modern meat substitutes cause more emissions than traditional meat. It's good in the "not harming animals" sense, but the environmental benefits are not really there.

**u/TougherThanAsimov** • 1 points • 2025-08-16 19:23 UTC

Okay, I actually might need you to gimme a quick rundown as to why. I mean, I'm pretty sure the FCR of cattle is at least 12 lbs of feed to 1 lb of meat, \*and that's live weight.\* The FCR numbers look worse for dressed carcasses.

So what are the logistics issues for fake meat? Are we using fancy plant products like pomegranate or something?

**u/printmyplastic** • 1 points • 2025-08-16 20:59 UTC

>So what are the logistics issues for fake meat? Are we using fancy plant products like pomegranate or something?

<https://www.ucdavis.edu/food/news/lab-grown-meat-carbon-footprint-worse-beef>

Lab-grown meat, which is cultured from animal cells, is often thought to be more environmentally friendly than beef because it's predicted to need less land, water and greenhouse gases than raising cattle. But in a preprint, not yet peer-reviewed, researchers at the University of California, Davis, have found that lab-grown or "cultivated" meat's environmental impact is likely to be "orders of magnitude" higher than retail beef based on current and near-term production methods.

Researchers conducted a life-cycle assessment of the energy needed and greenhouse gases emitted in all stages of production and compared that with beef. One of the current challenges with lab-grown meat is the use of highly refined or purified growth media, the ingredients needed to help animal cells multiply. Currently, this method is similar to the biotechnology used to make pharmaceuticals. This sets up a critical question for cultured meat production: Is it a pharmaceutical product or a food product?

"If companies are having to purify growth media to pharmaceutical levels, it uses more resources, which then increases global warming potential," said lead author and doctoral graduate Derrick Risner, UC Davis Department of Food Science and Technology. "If this product continues to be produced using the "pharma" approach, it's going to be worse for the environment and more expensive than conventional beef production."

The scientists defined the global warming potential as the carbon dioxide equivalents emitted for each kilogram of meat produced. The study found that the global warming potential of lab-based meat using these purified media is four to 25 times greater than the average for retail beef.

**u/TougherThanAsimov** • 1 points • 2025-08-16 21:49 UTC

Well, all of that *\*does\** track. But when I said fake meat, I meant skipping the middleman entirely and using synthesized protein and plant-based ingredients. Assuming we could get convincing replicas of ground meat via *\*\*dare I say\*\** a vegan option, what about that?

(This is useful though; I had a feeling that the sci-fi idea of growing meat in a vat was impractical.)

**u/printmyplastic** • 1 points • 2025-08-16 22:15 UTC

>Well, all of that *\*does\** track. But when I said fake meat, I meant skipping the middleman entirely and using synthesized protein and plant-based ingredients. Assuming we could get convincing replicas of ground meat via *\*\*dare I say\*\** a vegan option, what about that?

>(This is useful though; I had a feeling that the sci-fi idea of growing meat in a vat was impractical.)

I mean, this is synthesized protein. The way we do it now is crazy energy intensive. Doesn't mean it will always be that way. I'm not an expert on the topic, but I was under the impression that these synthetic meat companies were operating under the "let's disrupt things now and figure out how to make money later" tech business model. If they made an eco friendly meat substitute, sign me up; meat is delicious. I actually got one of those impossible burgers for the first time in forever today, I'm just not going to kid myself that it's saving the environment. It's also kind of funny that a lot of these products are deliberately not marketed as vegan because people would assume they taste bad if they were.

**u/Brilliant\_Goat\_2361** • 1 points • 2025-08-16 13:46 UTC

Thank you for the insight!

**u/QuantumModulus** • 1 points • 2025-08-16 04:39 UTC

I barely eat red meat anymore, and the fake meat is plenty tasty enough for me already honestly. The limiting factor for me is price, and I expect that to flip soon. Seeing reports that beef prices are starting to rise now because of droughts too.